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To cite this article: Jianbo Tang, Min Deng, Ju Peng, Zhiyuan Hu, Xingxiang Jiang, Jianbing Xiang, Xia Ning & Wenzhe Zhao (01 Sep 2025): How to describe spatiotemporal patterns of moving objects: a new classification framework, International Journal of Geographical Information Science, DOI: [10.1080/13658816.2025.2548819](https://doi.org/10.1080/13658816.2025.2548819)

To link to this article: <https://doi.org/10.1080/13658816.2025.2548819>



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RESEARCH ARTICLE



How to describe spatiotemporal patterns of moving objects: a new classification framework

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ABSTRACT

Rapid advances in positioning and monitoring technologies have significantly enhanced the ability to track dynamic moving objects for studying movement patterns. As an emerging field in spatio-temporal data mining, a well-established taxonomy of movement patterns facilitates tasks that mine movement patterns. In this study, we proposed a novel classification framework with the 5W1H1R (who, what, when, where, why, how, and relationships) principle and a bottom-up multi-level cognitive model to support the taxonomy of movement patterns. Guided by first principles thinking and combinatorics, we differentiated basic patterns categorized along spatial, temporal, and motion attribute dimensions from compound patterns composed of these basic patterns. We summarized five key constraints to refine movement patterns over recently studied pattern types. We validated our framework in three domains and compared it with four existing frameworks in five aspects. The results demonstrate the broader coverage, extensibility, and adaptability of the framework. Our classification framework can adapt to various moving objects, application domains, and movement data at different scales and resolutions. It can serve as a conceptual and ontological foundation for guiding the mining and analysis of movement patterns and, especially, for developing models to detect multimodal movement patterns.

ARTICLE HISTORY

Received 12 February 2025
Accepted 13 August 2025

KEYWORDS

Movement pattern; moving object; spatiotemporal patterns; classification framework; movement data

1. Introduction

The rapid evolution of the integration of monitoring technologies across space, air, and ground has revolutionized our ability to observe the Earth (Bitao 2022, Tampakis *et al.* 2022), enhancing the perception of all types of geographical elements, especially various dynamic moving objects. This has provided unprecedented opportunities to study the spatiotemporal distributions, interactions, and evolutionary trends of objects in geographical space (Atluri *et al.* 2019). Previous research has primarily focused on diverse static geographic elements, such as land use patterns (Smith and

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Supplemental data for this article can be accessed online at <https://doi.org/10.1080/13658816.2025.2548819>.

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Raemaekers 1998, Liang *et al.* 2021), urban functional area distributions and interactions (Liu *et al.* 2025), and road traffic flow predictions (Deng *et al.* 2023). Empowered by advances in real-time tracking technologies, researchers are focusing on the movement regularities of various dynamic moving objects. Moving objects are defined as those whose spatial locations or geometric attributes change over time (Dodge *et al.* 2008), including various visible geographical entities (e.g. humans, vehicles, and animals) and dynamic natural phenomena, such as hurricanes and spreading epidemics.

Over the past two decades, a key focus of movement-related research has been to investigate the spatial and temporal distributions and any interesting relationships between moving objects (i.e. movement patterns) (Laube and Imfeld 2002, Dodge *et al.* 2008, Tritsarolis *et al.* 2021, Francia *et al.* 2024). Similar movement patterns can be observed in moving objects from diverse domains, although they may reflect different behavioral semantics (i.e. domain-specific interpretations or intentions). For example, the convergence pattern may indicate foraging behavior in animals, but it may also indicate traffic congestion caused by vehicles in urban traffic systems. To date, various movement patterns have been defined and identified. However, most studies have focused on specific movement patterns linked to particular behavioral semantics within a single domain, overlooking similarities in movement patterns across diverse contexts. This limits the development of more general pattern mining algorithms capable of recognizing similar movement patterns across different moving objects, or even diverse movement patterns. This highlights the necessity of developing a unified classification framework that transcends specific application domains and moving object types to cognitively define and categorize these movement patterns. Such a framework not only provides ontological clarity by establishing clear categories of movement patterns but also enables the standardization of movement pattern analysis across disparate studies.

To date, several studies have attempted to classify movement patterns from different perspectives, such as relative motion (REMO) (Laube and Imfeld 2002), spatial and temporal characteristics (Dodge *et al.* 2008), the number of moving objects involved (Li 2014), and movement direction (Ji *et al.* 2016). However, different movement patterns can manifest at different spatiotemporal scales; some patterns only emerge at specific scales and become undetectable at others. A framework is critically needed to categorize these patterns in a way that adapts to variations across different scales and granularities. Therefore, this study proposes a novel classification framework that encompasses a wide range of movement patterns across multiple levels of abstraction. This framework aims to address four challenging questions: (1) What types of movement patterns exist, and how can they be systematically classified? (2) At what levels and dimensions can movement patterns be cognized and categorized? (3) At what spatiotemporal scales or granularities might these patterns manifest? (4) How generalizable is the proposed classification framework across different application domains? The contributions of this study are as follows:

- A novel ontology is introduced to conceptualize the movement patterns of moving objects from seven dimensions, and a multi-level model is developed to cognize these patterns.

- A systematic and extensible taxonomy of the movement patterns is proposed, which provides formal definitions and illustrative diagrams of the movement patterns.
- The applicability and generalizability of the proposed classification framework are validated through instantiations in three representative domains. In each case, we demonstrate the identification and interpretation of movement patterns within the unified classification scheme.
- We introduce the concept of the ‘mineability’ of movement patterns and discuss how the novel classification framework can be applied in practice.

The remainder of this paper is organized as follows: [Section 2](#) reviews related literature on movement patterns. [Section 3](#) introduces the conceptual foundations, and [Section 4](#) presents the constructed taxonomy of the movement patterns. [Section 5](#) summarizes common movement patterns and refines them using different constraints. [Section 6](#) presents the empirical validation of the proposed movement pattern classification framework. [Section 7](#) discusses the advantages and uses of our framework, along with the mineability of movement patterns. Finally, [Section 8](#) summarizes the study and provides directions for future research.

2. Related work

Movement pattern mining and analysis are crucial topics in contemporary movement data science (Gudmundsson *et al.* 2008, Dodge *et al.* 2020; Dodge 2021). A well-established classification framework provides the conceptual foundation that links an understanding of movement patterns to algorithm implementation. This section is organized into three parts: the cognition and classification of movement patterns, typical movement pattern mining methods, and existing limitations and challenges.

2.1. Movement pattern cognition and classification

Movement pattern cognition refers to the conceptual understanding of patterns in movement data. One of the earliest definitions of a pattern was provided by Fayyad *et al.* (1996) as an expression describing a subset of facts within a larger dataset that simplifies the representation of all facts rather than enumerating them individually. Subsequently, Laube and Imfeld (2002) introduced REMO, which characterizes movement patterns in a relative motion space using motion attributes such as speed, direction, and acceleration. Within the REMO framework, movement patterns are categorized into basic, spatial, and aggregate/segregate motion patterns (Laube *et al.* 2005b, Jeung *et al.* 2011). Subsequent research has primarily focused on movement patterns in geographic space and has attempted to formalize movement patterns and classify them systematically. Andrienko and Andrienko (2006) proposed four basic cognitive constructs for patterns: association, differentiation, arrangement, and distribution summary. These basic patterns were further specified for movement data and grouped into individual movement, momentary collective, and dynamic collective behaviors (Andrienko and Andrienko 2007). Building on existing studies, Dodge *et al.* (2008)

introduced a formal and systematic classification framework that distinguishes between generic patterns across domains and behavioral patterns with specific semantic meanings. Later studies cognized and categorized movement patterns based on the number of moving objects involved (Jeung *et al.* 2011, Li 2014) and the direction of collective movement (Ji *et al.* 2016). Andrienko *et al.* (2021) did not explicitly classify movement patterns, but introduced typical pattern types across various fields and constructed a theoretical model for pattern discovery. A comparison of the existing and proposed movement pattern classification frameworks is presented in Table 1.

2.2. Movement pattern mining

Building upon the cognition and definition of movement patterns, numerous patterns, whether derived from established or newly defined classification frameworks, have been identified and analyzed in diverse applications. Researchers have developed various algorithms to mine patterns from real-world movement data (Long and Nelson 2013, Gudmundsson *et al.* 2012, Dodge *et al.* 2020), which can be broadly divided into four categories. Among these methods, visual analytics provides an intuitive and interactive means for revealing potential movement patterns (Andrienko and Andrienko 2006, 2007, 2013). It is useful in exploratory analysis and guiding the visualization of movement patterns but may struggle with scalability and visual clutter when applied to large-scale or dense data. REMO-based mining identifies REMO patterns by matching predefined templates using motion attributes (Laube and Imfeld 2002, Laube *et al.* 2005a, Chavoshi *et al.* 2014, Ban and Kim 2023). Although extensions have been made to incorporate spatial proximity (Laube *et al.* 2005b), they still face challenges in defining complex or composite templates.

With advances in spatiotemporal data mining, various techniques have been adopted for movement pattern discovery, including clustering (Yuan *et al.* 2017a), association rule mining (Sridhar and Shanker 2014), and anomaly detection (Li *et al.* 2011, Shi *et al.* 2021). For example, Kalnis *et al.* (2005) utilized density clustering to detect spatially proximate groups of moving objects at each timestamp, connecting these clusters across consecutive timestamps based on the Jaccard index to identify moving clusters. Jeung *et al.* (2010) used density clustering to identify a set of fixed objects that travelled together. Peng *et al.* (2025) proposed a two-stage trajectory clustering method for detecting frequent movement paths for various moving objects. More recently, machine learning has emerged as a promising tool for movement pattern discovery. Yang and Gidófalvi (2020) converted trajectories into image representations and used a convolutional neural network to classify dominant movement patterns.

2.3. Critical analysis of movement pattern research

From the above review, we can discern that the accurate identification and effective visualization of movement patterns fundamentally depend on a clear conceptualization and definition of these patterns. Three key properties of movement patterns must be considered when establishing a classification framework for movement patterns. First,

Table 1. Comparisons of existing movement pattern classification frameworks and the proposed classification framework.

Research	Movement cognition	Classification scope	Movement pattern taxonomy	Flexibility/extensibility	Granularity/scale
Laube and Imfeld (2002)	Geometric primitives (motion attributes, REMO matrix)	Relative motion patterns	- Sequences: constance, turn - Incidents: concurrence, dispersion, bimodality - Interactions: trend-setter, and independent	Limited flexibility and extensibility	No explicit granularity/scale discussion
Andrienko and Andrienko (2006)	Conceptual grouping via visual-analytic semantics	General patterns (not limited to movement data)	Association, differentiation, arrangement, distribution	Systematic, flexible, and extensible	No explicit granularity/scale discussion
Andrienko and Andrienko (2007)	Movement characteristics of individuals and collectives at a single moment or over time	Individual and collective movement patterns specialized for movement data	- Similarity: constancy, repetition, - Differentiation: change, - Arrangement: trend, fluctuation, ...	Systematic, flexible, and extensible	Adaptable visualization scales, but no explicit scale/granularity discussion
Dodge <i>et al.</i> (2008)	Movement parameters and primitives across spatial, temporal, and spatiotemporal dimensions.	Generic (basic/compound) & behavioral patterns	Basic patterns: - Spatial: co-location in space, concentration, ... - Temporal: temporal sequence, periodicity, ... - Spatiotemporal: incidents, constancy, ... Compound patterns: repetition, convergence, ...	Systematic, flexible, and extensible	Discusses granularity/scale but lacks an explicit mechanism for movement patterns at different levels
Jeung <i>et al.</i> (2011)	Trajectory-based moving object relationships and relative motion	Trajectory patterns and relation motion patterns	Behavioral patterns: pursuit, foraging, migration, ... - Spatial/spatiotemporal trajectory patterns - Basic motion pattern: concurrence, trendsetter, ... - Spatial motion patterns: track, flock, leadership, - Aggregate/segregate motion patterns	Limited flexibility and extensibility	Granularity is discussed in terms of object grouping; no explicit spatiotemporal scale/granularity discussion.

(continued)

Table 1. Continued.

Research	Movement cognition	Classification scope	Movement pattern taxonomy	Flexibility/extensibility	Granularity/scale
Li (2014)	Object-centric perspective based on the number of moving objects	Trajectory and relation movement patterns for individuals and groups	- Individual period patterns - Pairwise movement patterns - Aggregate patterns - Co-directional movement patterns: moving cluster, convoy, swarm, ... - Directionally inconsistent movement patterns: convergence, divergence, ...	Extensible but limited flexibility	Object granularity only; no explicit spatiotemporal scale/granularity discussion
Ji <i>et al.</i> (2016)	Direction-similarity in groups	Collective movement patterns	Directionally inconsistent movement patterns: convergence, divergence, ...	Extensible but limited flexibility	No explicit scale/granularity discussion
The proposed	5W1H1R principle and multi-level cognition of movement patterns	Both relative motion patterns and movement patterns for individuals and groups at different levels	Basic patterns: - Spatial: co-location in space, concentration in space, anomaly in space, ... - Temporal: synchronization, precede, lag, - Motion attribute: similarity, difference Compound patterns: co-location in space and time, moving cluster, convergence, divergence, ...	Systematic, flexible, and extensible	An explicit multi-level mechanism for movement patterns, which is adaptive to movement data with different scales and granularities

* Note: Movement patterns in Table 1 refer to the patterns in geographical space, and relative motion patterns refer to the patterns in relative space.

movement patterns are diverse (Andrienko and Andrienko 2006), and some may be complex and difficult to detect. This complexity depends on the inherent pattern characteristics, data quality (e.g. signal-to-noise ratio and data resolution), and the specific feature extraction and detection algorithms employed. Therefore, it is necessary to clarify potential movement pattern types and their characteristic attributes. Second, movement patterns, even those of the same type, can occur at different spatiotemporal scales and granularities, which vary according to the application domain (Dodge *et al.* 2008). For specific domains, movement patterns are scale-dependent, with different patterns emerging at specific scales and granularities. Third, movement patterns are inherently hierarchical and decomposed (Gaidon *et al.* 2014). This means that complex patterns can be broken down into simpler sub-patterns, whereas simpler patterns can be combined to form more complex patterns or refined by introducing additional constraints.

However, most existing movement pattern classification frameworks do not adequately consider these non-negligible properties. Consequently, they fail to incorporate the movement patterns proposed in recent studies owing to the lack of a more explicit, extended classification mechanism. Among prior efforts, Dodge *et al.* (2008) made a significant contribution to the systematic classification of movement patterns. Their framework builds upon earlier fundamental studies (Laube and Imfeld 2002, Andrienko and Andrienko 2006, 2007) by synthesizing existing patterns and organizing them from spatiotemporal perspectives. It is also the most closely related to our work and serves as a key reference for our classification framework. The most significant difference between our study and that of Dodge is that we propose two cognition models to guide the classification process of movement patterns. Our approach emphasizes a more foundational decomposition and systematic combination of movement patterns at different levels, making our framework more systematic and extensible.

3. Conceptual foundations: the 5W1H1R ontology and multi-level cognitive model

Movements are driven by the internal intentions of moving objects and shaped by their interactions with external environments (other objects, spatial contexts, and temporal constraints). Movement patterns exhibited by moving objects are inherently diverse, scale-dependent, and hierarchically decomposable. A comprehensive cognition of moving objects and their movements provides a foundation for developing a systematic and hierarchical movement pattern classification framework. This study proposes a conceptual framework consisting of (1) a who, what, when, where, why, how, and relationships (5W1H1R) ontology for characterizing movement patterns, and (2) a multi-level cognitive model that is scale-adaptive for capturing movement patterns at different levels. These conceptual tools can guide the classification of movement patterns, the formal feature descriptions, and the design of pattern mining algorithms.

3.1. 5W1H1R ontology

The who, what, when, where, why, and how (5W1H) principle, originating from journalism, and the geographic information six-element models proposed by Lu *et al.* (2017) are two

Table 2. Descriptions of the dimension in the proposed 5W1H1R model for movement pattern cognition.

Cognitive dimension	Description
When	Temporal information of movement (e.g. start/end time, duration, frequency, periodicity).
Where	Spatial context of movement (e.g. origin/destination points, trajectories, sequences, routes, spatial extent, geographic regions).
Who	Identity and characteristics of moving objects, such as object ID, object type, physical attributes, and group/individual distinctions.
What	Types of movement patterns (e.g. moving cluster, dispersion, encounter, co-location, encounter)
Why	Behavioral semantics or motivations behind movement (e.g. commuting, migration, resource search)
How	Movement mechanisms, modes, or strategies, such as motion attributes (e.g. speed, direction, and acceleration), means of transport (e.g. walking, riding), and movement evolutions.
Relationship	Spatial, temporal, attribute, and semantic relationships between moving objects or between moving objects and geographical environments. Includes: • Spatial relations: distance, direction, topological relationships (e.g. adjacency, containment), • Temporal relations: simultaneity, sequence, delay, overlap, etc. • Attribute relations: similarity, difference, etc. • Semantic relations: social ties, functional associations, behavioral independence

important ontological approaches for modeling events or geographical phenomena. The 5W1H schema, which focuses on six basic components (when, where, who, why, what, and how), is a general-purpose schema for describing events. However, it lacks an explicit characterization of the relational dimension, which is essential for modeling group movement behaviors (e.g. interactions and coordination). In contrast, the geographic six-element model not only incorporates spatiotemporal positioning and attributes but also emphasizes the relationships and evolutionary processes of geographic elements. This model supports a more process-oriented understanding of geographic phenomena, but is less focused on the behavioral aspects of movement pattern cognition.

Drawing inspiration from these two influential models, this study proposes the 5W1H1R model, which extends the 5W1H schema by incorporating relationships (R) as a core dimension (Table 2). The addition of ‘relationship’ captures the spatial, temporal, attribute, or semantic interactions between moving objects or between moving objects and their environments. This enhances the expressiveness and relational awareness of prior movement conceptual frameworks (Dodge *et al.* 2008, Andrienko *et al.* 2011). The definitions of each dimension are further refined and strengthened to tailor the 5W1H1R model to movement pattern cognition. This model serves as an ontological foundation for describing the movements of moving objects in diverse contexts and supports the structured extraction and formalization of movement pattern features. A comparison between existing Geographic Information System (GIS) ontology models and the 5W1H1R model is presented in Table 3. By combining Tables 2 and 3, it can be observed that the proposed 5W1H1R model offers the best support for movement pattern cognition with more comprehensive dimensions and targeted descriptions. This enables researchers to interpret movement patterns not only using spatiotemporal properties or statistical descriptors, but also in terms of intentions, processes, and results, thereby providing a richer analytical perspective.

Table 3. Comparison of existing GIS ontology models and the proposed 5W1H1R model.

Ontology model	Core dimensions/elements	Support for movement			Limitations
		Focus area	Support for movement patterns	Strengths	
SW1H (Journalism)	When, Where, Who, What, Why, How	Event description	Strong but partial	General-purpose, intuitive structure	Lacks relational modeling
Time geography (Hägerstrand 1970)	Time, space (path, constraints)	Spatiotemporal process and events	Strong but partial	Strong in visualizing temporal constraints	Weak in semantic and relational modeling
Dodge <i>et al.</i> (2008)	Movement parameters (spatial, temporal, spatiotemporal)	Movement pattern modeling	Strong but partial	Emphasizes the conceptualization of movement	Limited higher-level cognition
Andrienko <i>et al.</i> (2011)	Space, time, moving objects, spatiotemporal context, and relations among elements	Movement, cognition, and analysis	Strong but partial	Emphasizes the conceptualization of movement	Overlooks behavioral semantics (i.e. Why/How)
Geo-ontologies (Fonseca and Câmara 2009)	Concept (name, definition, attribution), spatial relations, semantic relations	Geographical information modeling	Partial (static features)	Formalizes geographic concepts and relationships	Weak in temporal dynamics and behavioral modeling
Geographic six-element model (Lv <i>et al.</i> 2018)	Spatial positioning, geometric form, attributes, relationships, evolution process, and semantic description	Geographical objects or phenomena modeling	Strong and comprehensive	Rich in spatial semantics, evolution, and relationships	Overlooks behavioral semantics (i.e. Why/How)
The proposed 5W1H1R model	When, Where, Who, What, Why, How, Relationship	Movement pattern cognition and classification	Strong and comprehensive	Unified movement conceptualization and cognition	Required detailed data and domain knowledge for full implementation

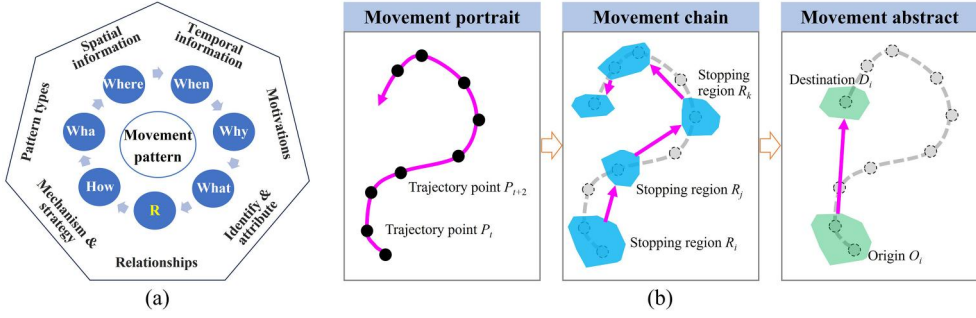


Figure 1. Cognition of movement patterns of (a) the 5W1H1R model and (b) the multi-level cognitive model.

3.2. Multi-level cognitive model for movement patterns

The 5W1H1R principle emphasizes the ontological characterization of movement patterns through multiple dimensions at specific spatiotemporal scales and object levels. Beyond the distinction between individual and collective movement patterns, the types of movement patterns that can be meaningfully recognized are inherently influenced by spatiotemporal scale and data granularity. Movement patterns can emerge at different scales and granularities. To address this, we propose a multi-level cognitive model that organizes the understanding of movement patterns from different levels of abstraction rather than fixed spatiotemporal scales. As shown in Figure 1(b), this hierarchically progressive model consists of three levels:

- **Micro atomic level:** This level focuses on the finest-grained movement observations (i.e. minimum sampling units), typically at the trajectory point level with relatively high temporal frequency and fine spatial granularity. It captures atomic movement features, such as instantaneous position, speed, direction, acceleration, and curvature, over short time intervals. This level supports the construction of detailed *movement portraits* of moving objects.
- **Meso episodic level:** This level abstracts fine-grained movements into meaningful episodes (e.g. stops and moves) or trajectory segments. It captures mid-level behavioral features, such as stop durations, segment lengths, dominant directions, and transitions between activity states, occurring over medium temporal windows and spatial extents, in contrast to the atomic level. This level supports the construction of *movement chains* that reflect high-level behavior of moving objects.
- **Macro flow level:** This level disregards fine-grained movement details (e.g. exact paths) and emphasizes the overall movement regularities of moving objects. Compared to atomic and episodic levels, it operates on relatively extended temporal durations and broader spatial extents and captures macro features, such as origin–destination trends and regional flows. This level supports the construction of *movement abstracts* for moving objects, which is helpful for analyzing aggregated movement behaviors.

The micro atomic, meso episodic, and macro flow levels represent three distinct abstractions of movement, ranging from fine-grained trajectory points to

coarse-grained origin–destination flows. At each level, movement features can be cognized and described using the 5W1H1R model. The key advantage of the proposed multi-level cognition model is that it relies on the abstract representation of movement data at different levels rather than on a fixed spatiotemporal scale or data granularity. The ‘level’ of movement cognition is not determined by the absolute sampling interval or spatial extent but by the movement representation within a given domain. This enables the multi-level model to accommodate different data resolutions and tracking durations, thereby adapting it to various types of moving objects and observation conditions.

This study provides a guideline for distinguishing movement patterns at different levels: patterns observed at the trajectory point level are generally categorized as the micro atomic level, patterns extracted based on stop–move episodes or trajectory segment states correspond to the meso episodic level, and patterns discovered based on aggregated flows represent the macro flow level. For example, although pedestrian trajectories recorded at 1-second intervals and vessel trajectories recorded every 10 minutes differ in temporal resolution, both reflect atomic-level movement behaviors because they capture fine-grained movement details relevant to their respective domains.

3.3. Representation of the movement data of moving objects

We introduce different representations of movement data, corresponding to different levels of the multi-level cognitive model. This is crucial for understanding our cognitive model and selecting appropriate data representations to support the extraction of movement patterns. As shown in [Figure 2](#), there are five main methods for representing movement data: trajectory, trajectory segments, feature point sequences, flow, and spatiotemporal graphs. The first four correspond to increasing levels of movement abstraction and support various movement pattern mining tasks. The final method provides a flexible structure that can further represent the movement data of the first four methods.

- *Trajectory*: This preserves the highest temporal and spatial resolutions by recording the continuous movements of moving objects over time as sequences of sampling points. This representation corresponds to the atomic level and is suitable for fine-grained movement analysis.
- *Trajectory segment*: A trajectory segment is a subpart of a complete trajectory. It represents a continuous and relatively homogeneous movement phase (e.g. between two stops), corresponding to the episodic level. Trajectory segments can be extracted based on spatial, temporal, semantic, or information criteria, such as the minimum description length (Lee *et al.* 2007), and time or distance gaps.
- *Feature point sequence*: A feature point sequence is a high-level abstraction of movement at the meso episodic level, typically representing meaningful events or transitions in movement, such as stops, turns, and activity changes. These feature points are extracted from raw trajectories using stop detection algorithms, spatiotemporal thresholds (e.g. time gaps), semantic rules, etc.

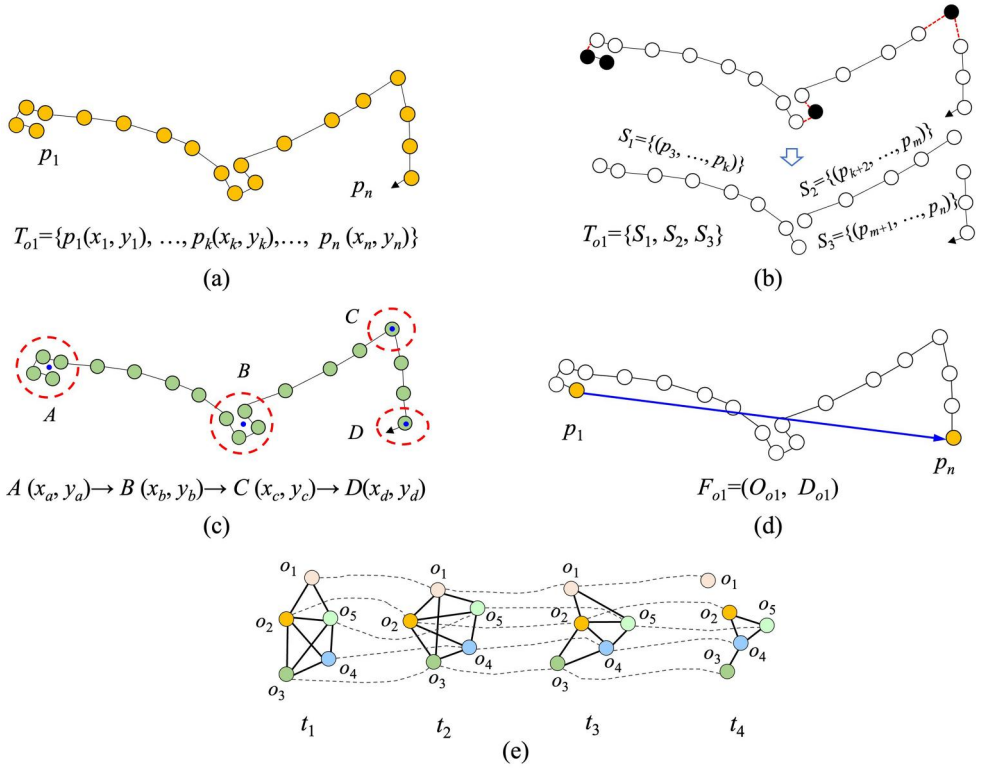


Figure 2. Representations of the movements of moving objects: (a) trajectory, (b) trajectory segment, (c) feature point sequence, (d) flow, and (e) spatiotemporal graph.

- *Flow*: This represents the highest-level abstraction of movement at the structural level. Typically, a flow is composed of an origin and destination (OD) pair, which is optionally associated with additional attributes, such as time interval and flow distance.
- *Spatiotemporal graph*: This is a flexible and expressive representation that abstracts moving objects as nodes and their relationships as edges (Tritsarolis *et al.* 2021). Formally, given a set of moving objects $O = \{o_1, \dots, o_n\}$, a graph snapshot at timestamp t_j is defined as $G_{tj} = (N, E_{tj})$, where $N = O$ (each node represents a moving object), and E_{tj} denotes the set of edges that capture the relationships among objects at t_j . The temporal evolution of movement is represented by a sequence of graph snapshots $G = \{G_{t1}, \dots, G_{tj}\}$, that is, a *spatiotemporal graph*. Edges between consecutive timestamps for the same moving object can be weighted by attributes, such as displacement or duration.

4. A novel classification framework of movement patterns

We first introduce the principles of movement pattern classification and provide an overview of the classification in Section 4.1. The taxonomy of movement patterns was developed based on prior studies, and various movement patterns are detailed in Sections 4.2 and 4.3. The schemas of movement patterns were developed based on

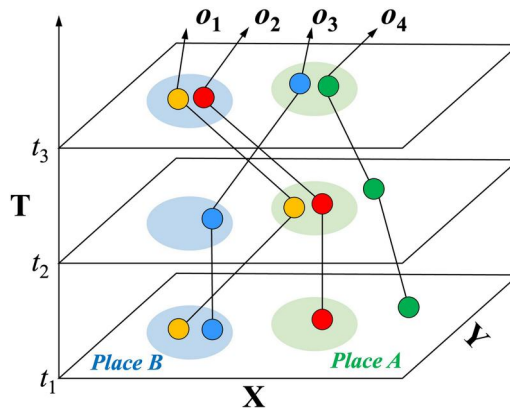


Figure 3. Schematic diagram of the movements of four moving objects (o_1 , o_2 , o_3 , and o_4) over three time intervals (t_1 to t_3). o_1 moves from place B to place A between t_1 and t_2 , then returns to place B from t_2 to t_3 . o_2 remains at place A from t_1 to t_2 and moves to place B between t_2 and t_3 . o_3 remains at place B from t_1 to t_2 and moves to place A from t_2 to t_3 . o_4 moves freely in the geographical space from t_1 to t_2 and arrives at place A at t_3 .

existing movement studies and formal pattern definitions, and were visually and expressively improved with reference to time geography (Hägerstrand 1970), particularly those involving spatiotemporal dimensions.

4.1. Overview of movement pattern classification

There are diverse movement patterns, some of which are complex. Guided by first-principles thinking, complex movement patterns can be decomposed into multiple relatively simple movement patterns. As illustrated in Figure 3, movement is inherently embedded in specific spatiotemporal contexts and is characterized by various motion attributes (e.g. speed and direction). The three dimensions of space, time, and motion are fundamental considerations for decomposing complex patterns. In addition, in our proposed cognitive model, movement patterns were cognized at three distinct levels corresponding to hierarchically progressive abstractions of movements. Based on this, we propose two principles for classifying movement patterns.

First, we decomposed movement patterns into basic patterns along the three dimensions of spatial, temporal, and motion attributes. Basic patterns describe the movements of moving objects from only a single dimension and serve as building blocks for relatively more complex patterns involving two or more basic patterns. At each dimension, the classification of basic patterns was based on the relationships among moving objects or their movement elements, ensuring that all possible variations within that dimension were considered. Following combinatorial principles, two or more basic patterns from two or more dimensions were then combined to form compound patterns. Second, we classified movement patterns across three cognitive levels of movement abstraction. At each level, movement patterns were first categorized along three fundamental dimensions to form basic patterns, which were then combined into compound patterns.

These two guidelines ensure that the proposed classification framework adapts to different moving objects, spatiotemporal scales, and domains. By addressing fundamental dimensions and hierarchical movement abstraction levels, the proposed framework is both flexible and extensible. It can cover various existing and emerging movement patterns. An overview of the classification of movement patterns is presented in Figure 4. Although the movement patterns included in this study may not encompass all possible types encountered in practice, the classification principles and framework we established are scalable and domain-adaptable, providing a strong foundation for theoretical extension and various practical applications.

4.2. Basic movement patterns

Basic movement patterns are categorized along three fundamental dimensions (spatial, temporal, and motion attributes). To formalize these patterns, we first introduce some notation. Let $O = \{o_1, o_2, \dots, o_n\}$ be a set of n moving objects, and let $T = \{T_1, T_2, \dots, T_n\}$ be their corresponding trajectories. Each trajectory $T_i = \{(x_i^1, y_i^1, t_1), (x_i^2, y_i^2, t_2), \dots, (x_i^k, y_i^k, t_k)\}$ is represented by k timestamped points.

1. Basic spatial movement patterns

Basic spatial patterns focus on the spatial sequences of moving objects or the location-based relationships among them, which are classified into five fundamental types.

- *Spatial sequence*: This refers to a set of places or regions visited by one or more moving objects without including temporal details, such as visit times and durations. As illustrated in Figure 5(a), it corresponds to the movement pattern at the episodic-level cognition, emphasizing important locations rather than raw trajectory points. Formally, the spatial sequence of the moving object o_i can be defined as:

$$S_{oi} = \{r_1, r_2, \dots, r_m\} \quad (1)$$

where $r_i \in S_{oi}$ denotes the region or place extracted from the raw trajectory using stop-point detection or semantic mapping methods such as density clustering.

- *Co-location in space*: This refers to the presence of two or more moving objects in some common locations or regions, regardless of temporal information. This pattern reveals shared spatial behaviors, such as a series of attractions visited by different tourists, as shown in Figure 5(b). Let $S_{oi} = \{r_1^i, r_2^i, \dots, r_m^i\}$ and $S_{oj} = \{r_1^j, r_2^j, \dots, r_p^j\}$ be the sets of visits of moving objects o_i and o_j , respectively. The co-location in space can be formalized as follows:

$$|S_{oi} \cap S_{oj}| \geq \theta \quad (2)$$

where $|S_{oi} \cap S_{oj}|$ denotes the number of common locations between o_i and o_j , and θ is the minimum threshold of common locations in space.

- *Concentration in space*: This refers to the spatial aggregation of multiple moving objects (generally more than two objects) or their movements, which typically form dense spatial clusters. According to different levels, spatial concentration

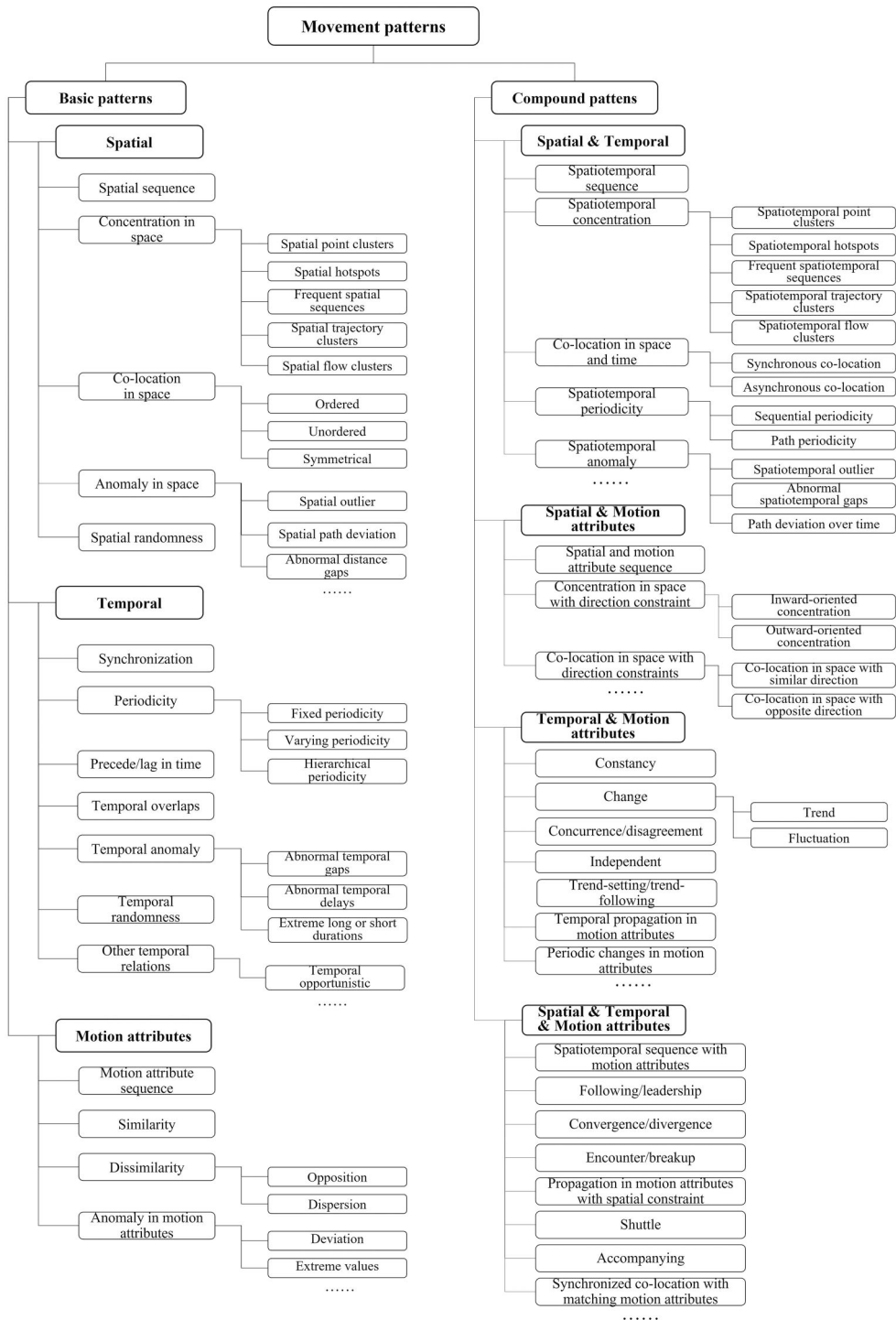


Figure 4. Overall taxonomy of movement patterns.

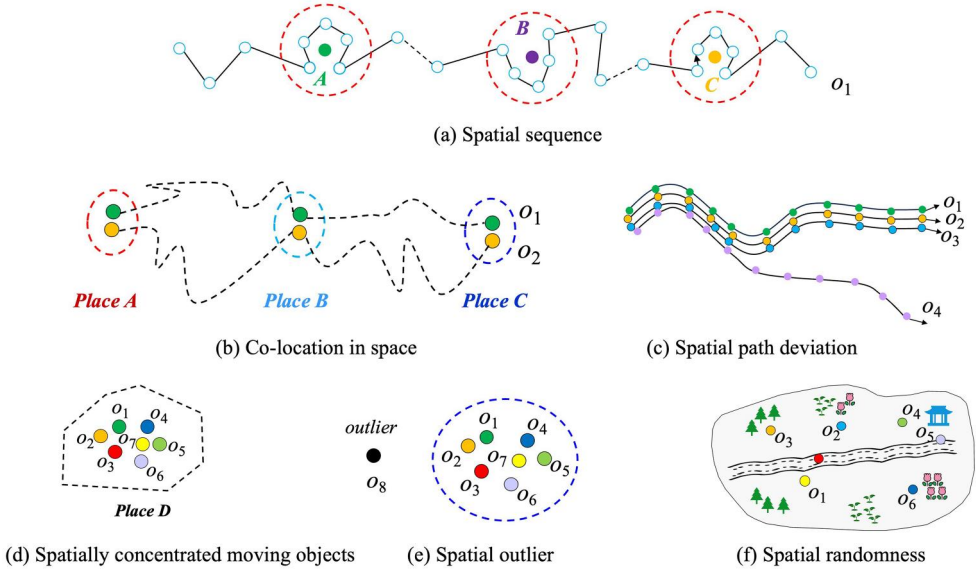


Figure 5. Schematic diagrams of basic spatial patterns (circles of different colors represent different moving objects).

manifests as spatial object clusters, spatial hotspots, frequent spatial sequences, spatial trajectory clusters, and spatial flow clusters. As illustrated in Figure 5(c), a spatial concentration of seven moving objects $C \in O$ can be formalized (not uniquely) as:

$$\forall o_i, o_j \in C, d(o_i, o_j) \leq d_r \quad (3)$$

where $d(o_i, o_j)$ is the spatial distance between o_i and o_j , and d_r is the maximum distance threshold.

- *Anomaly in space*: This refers to the spatial deviation of one or more moving objects or their movements from normal or historical patterns, or from those of other moving objects. Figure 5(d) depicts two common types of spatial anomalies: spatial outliers and spatial path deviations. For example, moving object o_i is considered a spatial outlier if:

$$\frac{1}{|O| - 1} \sum_{o_j \in O, o_j \neq o_i} d(o_i, o_j) > \delta \quad (4)$$

where δ is a predefined threshold indicating significant deviation.

- *Spatial randomness*: This refers to the independent or irregular spatial distribution of moving objects or their movements, where no discernible basic spatial patterns, as mentioned above, are observed. For instance, as shown in Figure 5(e), individuals may move randomly in a park, taking various paths for walking, jogging, or resting.

2. Basic temporal movement patterns

Basic temporal patterns focus on the temporal relationships of moving objects, aiming to describe how their movements are temporally connected, and are classified into seven basic types:

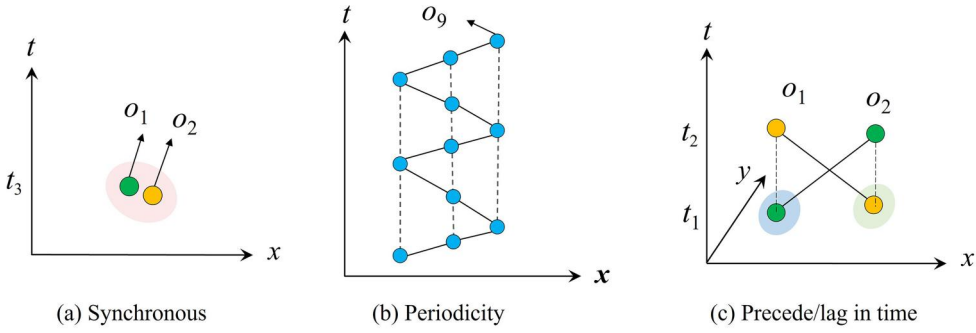


Figure 6. Schematic diagrams of movement patterns in the temporal dimension.

- *Synchronization*: This describes the simultaneous or temporally proximate movements of two or more moving objects (e.g. simultaneous starts/stops), as shown in [Figure 6\(a\)](#). This can be formalized as:

$$|t_{oi} - t_{oj}| \leq \delta_t \quad (5)$$

where t_{oi} and t_{oj} denote the timestamps of moving objects o_i and o_j , respectively. Two movement events of o_i and o_j are considered synchronized if their time difference falls within the temporal tolerance threshold δ_t . The value of δ_t is domain-specific. It can be user-defined based on the temporal resolution of movement data or learned adaptively based on temporal distribution statistics.

- *Periodicity*: This refers to the repeated occurrence of similar movements by one or more moving objects at regular time intervals, such as tidal movements and scheduled bus routes, as illustrated in [Figure 6\(b\)](#). The period can be fixed (strictly regular), variable (with flexible intervals), partial, global, or hierarchical. Let $M_{oi}(t)$ denote the movement of a moving object o_i at time $t \in \mathcal{T}$. It is periodic if:

$$\text{Sim} (M_{oi}(t), M_{oi}(t + kT)) < \varepsilon, \text{ for some } k \in \mathbb{Z} \quad (6)$$

where $\mathcal{T}$ is the temporal observation domain, T refers to the period, $\text{Sim} (.)$ denotes the movement similarity relation function, and ε is a temporal tolerance threshold within which temporal deviations are considered acceptable. Various temporal similarity measures (e.g. Dynamic Time Warping, autocorrelation, and Fourier analysis) can be used to quantify the periodicity of movement.

- *Precede/lag in time*: This captures the temporal ordering of movement events between two or more moving objects, where one or more objects precede or lag others over time. For example, as shown in [Figure 6\(c\)](#), object o_1 arrives at place A before o_2 (o_1 precedes o_2), but arrives at place B after o_2 (o_1 lags o_2). This precedence/lag relationship can be formalized as:

$$t_{oi} - t_{oj} > \delta_p \quad (7)$$

where t_{oi} and t_{oj} denote the timestamps at which the movement event occurs for moving objects o_i and o_j , respectively, and δ_p is a temporal threshold that defines the minimum significant precedence or following. In general, δ_t (a small time window) is smaller than δ_p (an apparent time gap). For example, delivery

trucks always arrive at the customer's location before the support vehicle within a time interval of 15–30 minutes. This exemplifies a precedence pattern. Two customers arrive at a certain store almost simultaneously with an interval of 5 seconds. This exemplifies a synchronization pattern.

- *Temporal overlap*: This refers to the overlap or intersection of the movement time intervals of two or more moving objects, regardless of whether their spatial positions coincide. The overlap can be partial or complete. Let $I_i = [t_s^i, t_e^i]$ and $I_j = [t_s^j, t_e^j]$ be the movement time intervals of objects o_i and o_j , respectively. Then, a temporal overlap exists if:

$$I_i \cap I_j \neq \emptyset \quad (8)$$

- *Temporal anomaly*: This refers to irregularities or deviations in the timing of the movements of moving objects relative to their historical/normal movements or other objects. These anomalies can have extremely long or short durations, abnormal temporal gaps, or delays, as illustrated in Figure 6(d).
- *Temporal relations*: This refers to any other possible temporal interactions among moving objects beyond the basic temporal patterns (Dodge *et al.* 2008). A typical and interesting example is temporal opportunistic movement, where the timing of the movement is not predetermined or planned but rather triggered by external opportunities, conditions, or events.
- *Temporal randomness*: This refers to the movement of one or more moving objects without any discernible distribution, regularity, or structure in a temporal dimension.

3. Basic motion attribute movement patterns

Basic motion attribute patterns focus on the motion attributes of moving objects (e.g. speed and direction) and are divided into four basic types based on REMO research:

- *Motion attribute sequence*: This refers to a sequence of motion attributes without spatiotemporal information. In REMO research, the motion attributes of multiple moving objects are represented using motion attribute matrices, such as the direction sequences shown in Figure 7 (Laube and Imfeld 2002).
- *Similarity*: Two or more moving objects exhibit the same or similar motion attributes, such as direction, speed, and acceleration. For example, a group of cyclists maintains a similar direction while riding, as shown in Figure 8(a).
- *Dissimilarity*: The motion attributes of a set of moving objects differ. As illustrated in Figure 8(b,c), there are two types of dissimilarity: dispersion and opposition.

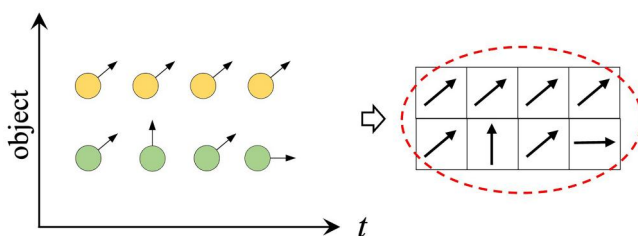


Figure 7. Motion attribute sequences of moving objects, illustrated using the direction attribute as an example.

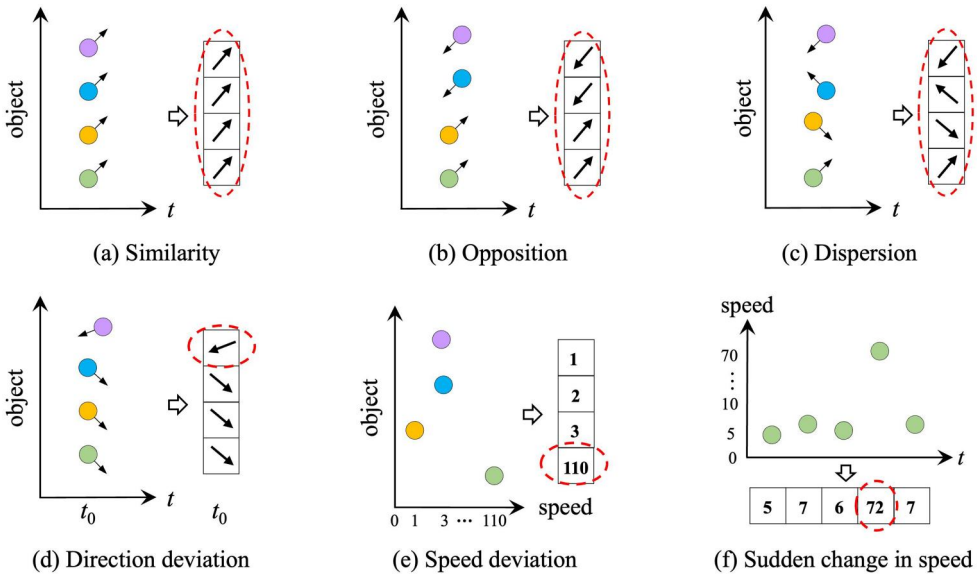


Figure 8. Schematic diagrams of movement patterns in the motion attribute dimension.

- *Anomaly in motion attributes:* This refers to irregularities or deviations in the motion attributes of one or more moving objects, regardless of their spatial locations or timing, such as extreme values or sudden changes (deviations) in motion attributes, as shown in Figure 8(d,e).

4.3. Compound movement patterns

Compound movement patterns are formed by the combination of two or more basic movement patterns across different dimensions. Based on this combination principle, we classified the compound patterns along four combination dimensions. While a full set of compound patterns can be theoretically derived by enumerating all basic combinations, this study focused on those most commonly observed and studied in real-world applications, owing to limited space. However, this does not imply that other compound patterns are unimportant; rather, they may be of interest in specific domains or contexts.

4.3.1. Compound movement patterns across spatial and motion attribute dimensions

The combination of basic movement patterns from the spatial and motion attribute dimensions can be understood from two perspectives. One is spatially constrained basic motion attribute patterns, and the other is basic spatial patterns with motion attributes that function as descriptors of the internal dynamics of moving objects. By combining basic spatial and motion attribute patterns, we can obtain over 20 compound patterns. Several meaningful patterns are illustrated in Figure 9.

- *Spatial and motion attribute sequence:* This refers to a set of places or regions visited by one or more moving objects associated with motion attribute information. This can be formally expressed as:

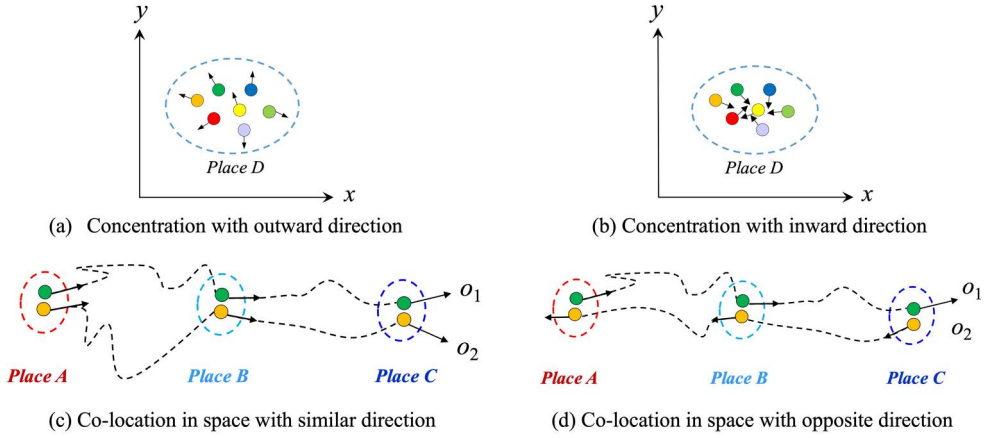


Figure 9. Schematic diagrams of typical compound patterns across spatial and motion attribute dimensions.

$$S_{oi} = \{(r_1, a_1), (r_2, a_2), \dots, (r_m, a_m)\} \quad (9)$$

where $(r_i, a_i) \in S_{oi}$ denotes a tuple of regions/places and motion attributes of moving object o_i .

- *Concentration in space with direction constraints:* This pattern extends spatial concentration by incorporating direction constraints. When moving objects in a concentrated area exhibit outward directions, this pattern is centrifugal; whereas when they exhibit inward directions, it is centripetal.
- *Co-location in space with direction constraints:* This pattern extends co-location in space by incorporating movement direction constraints. Moving objects in spatial co-locations may exhibit similar or opposite directions.

4.3.2. Compound movement patterns across temporal and motion attribute dimensions

The compound movement patterns across spatial and motion attribute dimensions can be derived not only from the combination of basic temporal patterns and motion attribute patterns but also from the evolutionary regularities of motion attributes over time. In this category, compound movement patterns capture the temporal dynamics of motion attributes. Among these patterns, we highlighted several representative types with significant practical relevance, as shown in Figure 10.

- *Constancy:* This pattern refers to the motion attributes (e.g. speed or direction) of moving objects that remain relatively stable or unchanged over a given time interval (Andrienko and Andrienko 2007).
- *Change:* This refers to the variation in motion attributes over time, which can be further divided into *trend/fluctuation* (Andrienko and Andrienko 2007). *Trend* denotes consistent or regular changes in motion attributes, whereas *fluctuation* represents irregular changes in motion attributes (Dodge et al. 2008).
- *Concurrence/disagreement:* *Concurrence* integrates synchronization and similarity patterns, indicating that multiple moving objects simultaneously exhibit similar

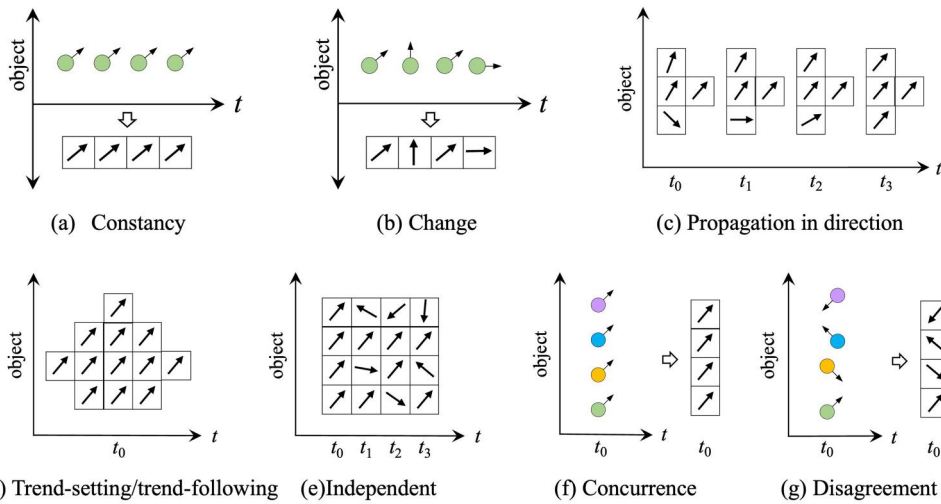


Figure 10. Schematic diagrams of typical compound patterns across temporal and motion attribute dimensions.

motion attributes (e.g. direction and speed). Disagreement combines synchronization and dissimilarity patterns, which indicate that moving objects display opposite or completely different motion attributes simultaneously.

- *Independent*: This refers to the persistence of dissimilarity in motion attributes among moving objects over time, indicating that their movements are independent of each other (Laube and Imfeld 2002)
- *Trend-setting/trend-following*: *Trend-setting* refers to the phenomenon in which the motion attributes of a moving object (termed the trend-setter) lead to those of other moving objects over time. In contrast, *trend-following* describes a situation in which the motion attributes of moving objects (termed trend-followers) follow the motion attributes of the trend-setter.
- *Temporal propagation in motion attributes*: This occurs when one or more moving objects begin to exhibit a specific motion attribute value (e.g. flying south or accelerating), and this behavior gradually spreads to other moving objects over time (Dodge *et al.* 2008). This pattern does not require moving objects to be close in space but highlights the cascading adoption of similar motion attributes.
- *Periodic changes in motion attributes*: This refers to the cyclical variations in motion attribute values at regular time intervals.

4.3.3. Compound movement patterns across spatial and temporal dimensions

Spatial and temporal dimensions are the two most crucial aspects in the study of movement patterns. The combination of basic spatial and temporal patterns can result in over 40 possible compound movement patterns. Some common patterns are listed below and shown in Figure 11:

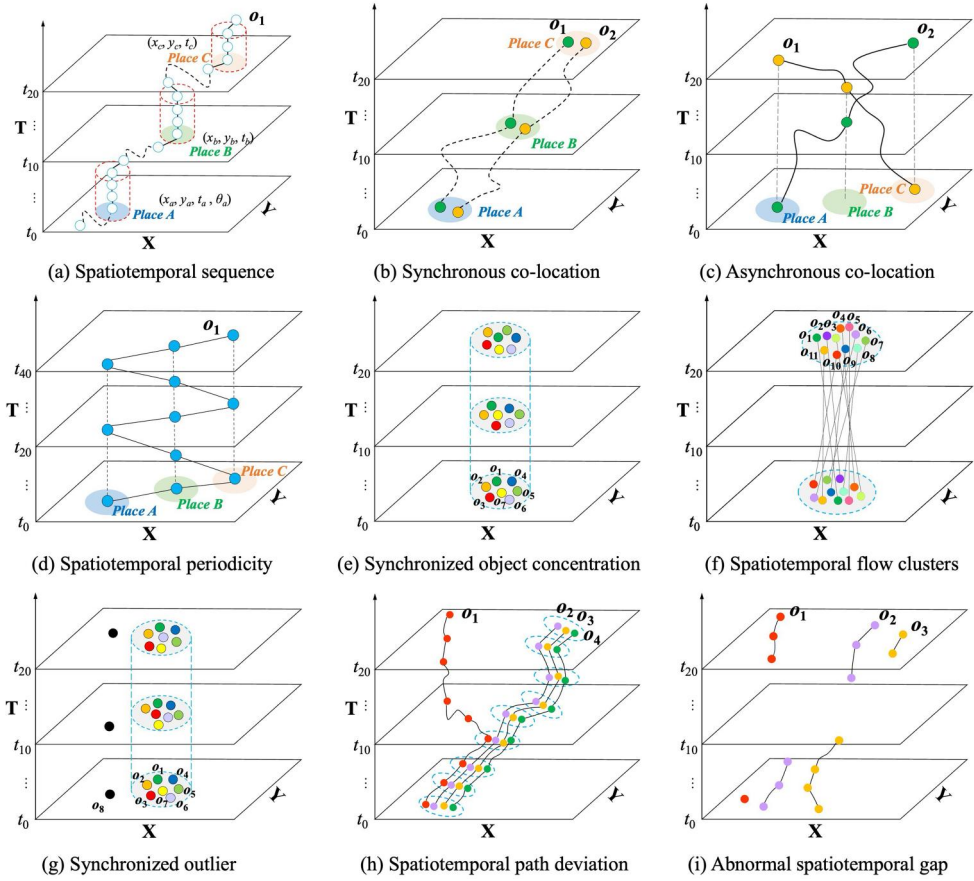


Figure 11. Schematic diagrams of typical compound movement patterns across spatial and temporal dimensions.

- *Spatiotemporal sequence*: A set of visited places/regions with temporal information that reveals the order or timing of the movement of moving objects between different locations.
- *Co-location in space and time*: This refers to the appearance of two or more moving objects in some common locations with temporal constraints (synchronous or asynchronous). Synchronous co-location means that objects arrive at the same location simultaneously or within a short time window. Asynchronous co-location indicates that objects visit shared locations at different times.
- *Spatiotemporal periodicity*: This refers to the regular repetition of movements across space and time, such as path and sequential periodicity (i.e. recurrence of paths or sequences at regular time intervals).
- *Spatiotemporal concentration*: This refers to the spatial aggregation of multiple moving objects or their movements with temporal constraints. This type of pattern includes spatiotemporal object clusters, hotspots, frequent spatiotemporal sequences, trajectory clusters, and flow clusters. These concentrations may be synchronous or asynchronous.

- *Spatiotemporal anomaly*: This extends spatial anomalies by incorporating temporal constraints, including spatiotemporal outliers, spatial path deviations over time, and abnormal spatiotemporal gaps.

4.3.4. Compound movement patterns across spatial, temporal, and motion attribute dimensions

The last combination considers the spatial, temporal, and motion attribute dimensions, yielding the most complex compound movement patterns. The basic patterns in any single dimension can explicitly assist in describing patterns involving the other two dimensions. Several commonly studied and observed movement patterns and their schematic diagrams (Figure 12) are as follows:

- *Spatiotemporal sequence with motion attributes*: A sequence of key points/regions of moving objects with temporal information and motion attributes.
- *Following/leadership*: This pattern extends the trend-setting/trend-following pattern by incorporating spatial constraints, where leaders and followers must maintain specific spatial distance relationships over time. For example, in a vehicle platoon, following vehicles maintain safe distances while matching the speeds of the preceding vehicle.
- *Convergence/Divergence*: *Convergence* describes multiple objects approaching a common spatial region from different directions, whereas *divergence* describes the opposite process, where objects disperse from a shared spatial region toward different directions. Our definition does not require the motion azimuth vectors of moving objects to intersect within a specific range or time window. This is applicable to movements restricted by the physical environment (e.g. road networks).
- *Accompanying*: A group of spatially proximate moving objects that move together in similar directions over time, such as a flock of birds.
- *Encounter/breakup*: An *encounter* refers to multiple moving objects arriving at the same location simultaneously, whereas a *breakup* describes their synchronous departure from a shared location (Dodge *et al.* 2008). An *encounter* can be viewed as the temporally constrained end stage of convergence, whereas a *breakup* marks the beginning of divergence. Both occur at a given moment or within brief time intervals.
- *Propagation with spatial constraint*: This pattern extends the propagation of motion attributes over time by imposing spatial constraints. As motion attributes spread among moving objects over time, they must maintain specific spatial relationships, such as remaining within a certain distance.
- *Shuttle*: A *shuttle* refers to the repetitive back-and-forth movement of one or more moving objects between two or more fixed locations, such as a ferry shuttling between islands. It generalizes the symmetry pattern (i.e. strict mirrored movement) defined by Andrienko and Andrienko (2007) by relaxing constraints on path, timing, and stop intervals, allowing for asymmetry in direction, route, or duration.

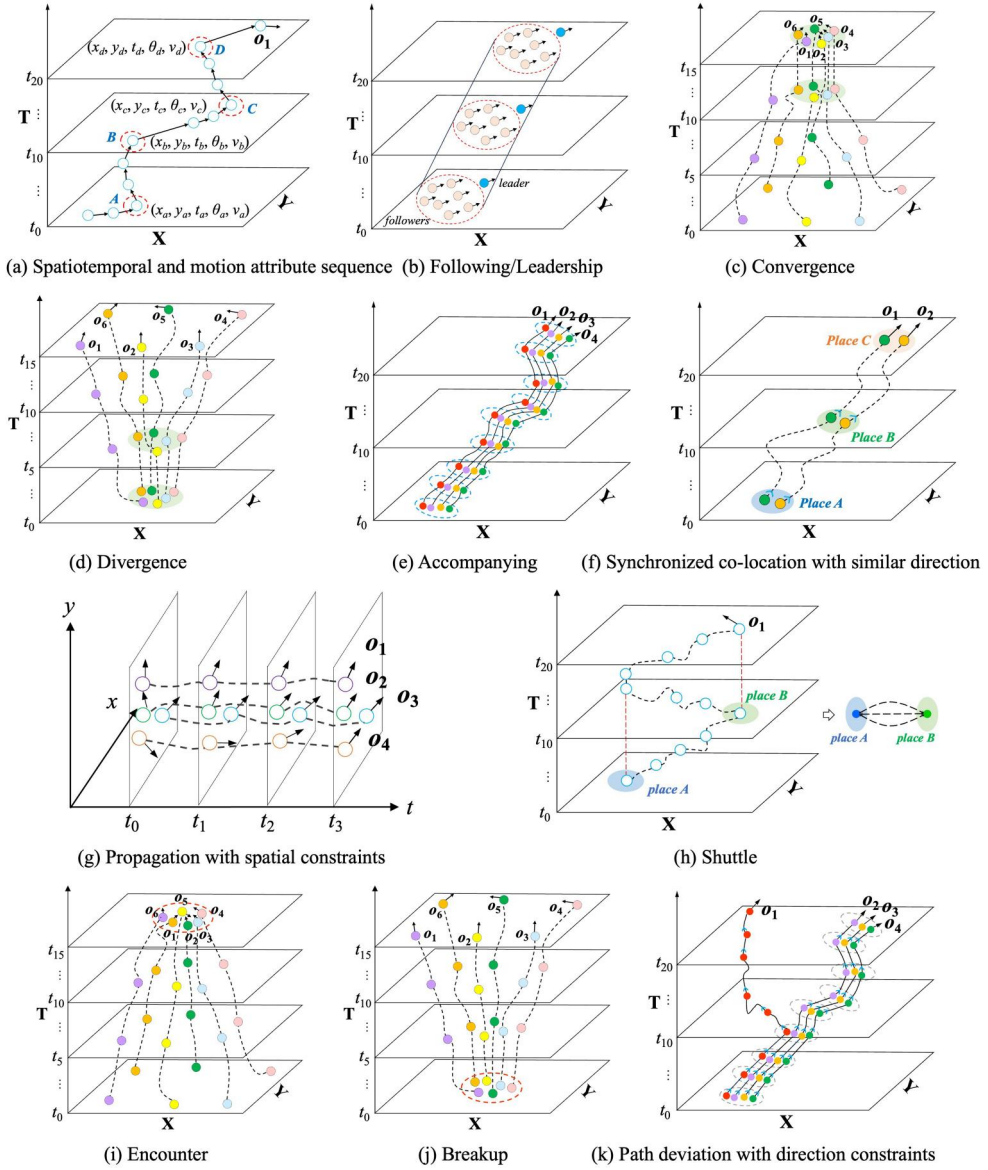


Figure 12. Schematic diagrams of typical compound patterns across spatial, temporal, and motion attribute dimensions.

- *Synchronized co-location with matching motion attributes:* This refers to two or more moving objects appearing in two or more common places or regions simultaneously, exhibiting similar motion attributes (e.g. speed and direction).
- *Spatial path deviation with motion attribute constraint:* This refers to the movement of one or more moving objects that significantly deviate from the collective, normal, or expected movements, with additional constraints on motion attributes, such as speed or direction.

5. Extension of movement patterns with constraints

5.1. Constraints for movement pattern extension

Each movement pattern type, particularly compound patterns, can be further extended by applying different constraints, which can yield movement patterns with practical application values but are not explicitly defined in the proposed classification. In this study, we summarized five key constraints that can be used to extend and customize movement patterns according to the proposed movement pattern classification framework.

1. Spatial constraints

- *Shape of spatial extent*: The spatial shape formed by movement patterns can be regular, fixed (e.g. fixed discs), or irregular (e.g. concave).
- *Variability of spatial extent*: The spatial range of the movement pattern was fixed (unchanged boundaries) or variable (expanding, contracting, or shifting).
- *Topological constraints*: This represents the topology of the spatial relationships of moving objects, such as containment (e.g. one group of objects moving in a specific area), adjacency (e.g. objects moving in neighboring regions), proximity, overlap, and connectivity.

2. Temporal constraints

- *Temporal persistence*: This represents whether the movement pattern persists for a minimum time threshold or occurs within a specific time window.
- *Temporal continuity*: This refers to whether the movement pattern must maintain a continuous presence (e.g. lasting for at least k continuous timestamps) or temporal gaps or breaks.
- *Temporal synchronization*: This refers to whether the movements of the members involved in the movement pattern are aligned in time (i.e. occur or change synchronously).
- *Temporal scope/structure*: The temporal scope of a movement pattern can be fixed, variable, local, global, or hierarchical. It is often used to refine periodic movements at various temporal resolutions.

3. Motion attribute constraints

- *Directional consistency*: Moving objects involved in the movement pattern maintain consistent directions or originate from different directions.
- *Speed or acceleration restriction*: This refers to whether the movement pattern requires matched speeds, accelerations, or other motion attribute values among the moving objects involved.

4. Moving object constraints

- *Minimum participation*: This refers to the minimum number of core moving objects that remain throughout the pattern and the minimum number of moving objects involved at any given timestamp.
- *Variability/dynamics of moving objects*: This refers to whether the moving objects involved in the movement pattern are fixed or varying (i.e. moving objects are allowed to leave or join).
- *Role constraints*: This refers to the role relationships between moving objects in the pattern (e.g. the leader–follower relationships). For example, wolves in different positions within a wolf pack may have different roles.

5. Geographical environment constraints

This constraint refers to the physical or semantic limitations imposed by the geographical environment in which moving objects move, such as road networks, spatial corridors, and natural barriers. These constraints vary according to the type of moving object. For example, the movements of pedestrians and vehicles are typically constrained by road networks, whereas ships navigate relatively freely but are restricted to waterways, and animals move more freely but are limited by natural barriers.

5.2. Applying constraints to existing typical movement patterns

Referring to the established movement pattern classification framework, we summarized eight common pattern types: accompanying, concentration, convergence, divergence, co-location, frequent, periodic, and abnormal movement patterns. Among these, we selected three typical patterns (i.e. accompanying, convergence, and periodic) to demonstrate the extension of existing movement patterns using the constraints described above.

5.2.1. Accompanying movement patterns

Accompanying movement patterns describe the behavior of a group of moving objects traveling together under certain spatial, temporal, and membership constraints. Depending on whether there is an evident leading moving object (i.e. the leader that moves in front of other moving objects (followers)) during the process of accompanying forward movement, the pattern can be divided into those with leaders and those without leaders (Andersson *et al.* 2008).

The accompanying pattern with leaders, also known as the following-leadership pattern, can be subdivided based on three criteria combined with the above constraints: (1) whether the spatial proximity between the leader and followers changes; (2) whether strict temporal continuity is required; and (3) whether the number of following and leading moving objects is fixed. The accompanying pattern without an obvious leader, in which all the involved moving objects share similar movements, is one of the most widely studied movement patterns. Typical patterns include flocks (Gudmundsson and Van Kreveld 2006; Benkert *et al.* 2008), moving clusters (Kalnis *et al.* 2005), convoys (Jeung *et al.* 2010; Liu *et al.* 2023), swarms (Li *et al.* 2010), traveling companions (Tang *et al.* 2014), platoons (Li *et al.* 2015; Ma *et al.* 2021), and their variants. Based on existing research, we summarized four criteria for refining this pattern: (1) whether strict temporal continuity is required; (2) whether the moving objects involved in the pattern must remain unchanged or are allowed to leave and join; (3) whether the spatial extent of the movement of moving objects is variable; and (4) whether the spatial shape must be fixed.

5.2.2. Convergence movement patterns

Convergence movement patterns describe the behavior of multiple moving objects approaching and concentrating within a specific spatial region (Jia *et al.* 2025). This pattern can be understood from two perspectives: (1) a certain number of moving

objects from dispersed locations converge at a particular destination; and (2) an increasing number of moving objects gradually gather together and continue to move as a group. To refine the convergence pattern, we summarize four criteria:

- Based on its different stages, this pattern can be divided into converging (approaching) and aggregation (post-formation) phases (Zhao *et al.* 2020). In group event modeling, the former generally corresponds to the fermentation stage of the group event, whereas the latter indicates the formation of the event.
- Whether the group size and members are required to remain fixed or may vary during the converging process (Hai *et al.* 2012, Guo *et al.* 2014).
- Whether the aggregation region is fixed or variable (Gudmundsson and Van Kreveld 2006, Zheng *et al.* 2013, Zhang *et al.* 2014).
- Whether strict temporal continuity of the pattern is required.

5.2.3. Periodic movement patterns

Periodic movement patterns refer to the recurring movements of moving objects at regular time intervals (Li *et al.* 2011, Li *et al.* 2012; Yuan *et al.* 2017b). To refine the periodic patterns, we summarize four key criteria: (1) whether the periodicity length is strictly fixed or exhibits slight variation; (2) whether the periodicity is observed locally or globally; (3) whether the spatial path in each cycle is required to be exactly the same or may vary; and (4) whether the periodicity structure is single-layered or exhibits a hierarchal form with multiple nested periodic layers.

6. Empirical validation of the proposed classification framework

To demonstrate the utility and generalizability of the proposed classification framework, we applied it to three real-world domains involving diverse moving objects from five aspects. We assessed its comprehensiveness by comparing it with four existing frameworks (Laube and Imfeld 2002, Andrienko and Andrienko 2007, Dodge *et al.* 2008, Li 2014), as listed in Table 1.

1. Movement data characteristics

The representative movement datasets from these domains are visualized in Figure 13. Table 4 presents their basic information and statistical characteristics. As observed, movement data from different domains differ significantly in terms of spatial extent, scale, and granularity. This poses challenges in the design of scalable movement pattern mining algorithms. In this context, the well-designed ontology of movement patterns, as provided by our framework, can offer a unified foundation for describing and comparing movement patterns across diverse datasets.

2. Factors influencing movements

Factors influencing the movements of moving objects are inherently domain-specific. In urban environments, human movement is driven by socioeconomic factors, such as work, education, and leisure, whereas animal movement is driven

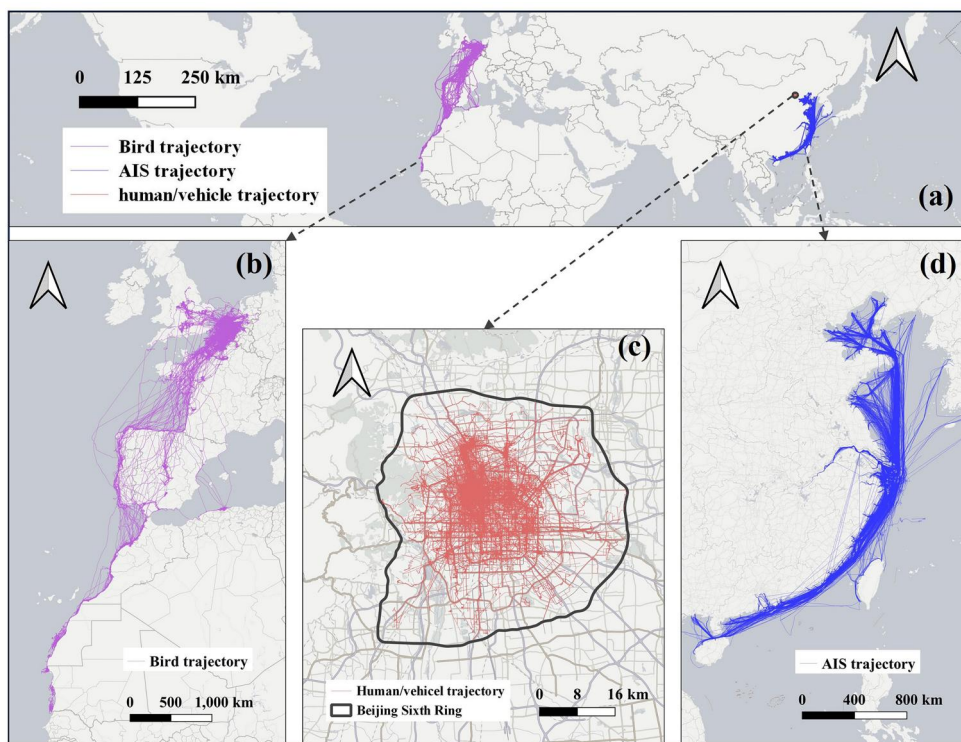


Figure 13. Representative movement datasets from different domains: (a) overview of all datasets, (b) bird trajectories, (c) human/vehicle trajectories, and (d) AIS trajectories.

by ecological or biological needs (e.g. foraging, migration, and breeding). Ship movement is influenced by navigational rules, goals, and the marine environment.

3. Cognition and interpretation of movement patterns

The proposed framework provides formal and comprehensive cognition of movement patterns supported by the 5W1H1R and multi-level models, which is lacking in baseline frameworks. Beyond spatiotemporal characterization, it supports semantic and contextual interpretation by associating observed movement patterns with domain-specific meanings through the ‘why’ and ‘how’ dimensions of the 5W1H1R model. In contrast, baseline frameworks primarily focus on the spatiotemporal aspects of movement patterns and lack explicit mechanisms for integrating interpretive semantics.

4. Spatiotemporal scale and granularity of movement patterns

Figure 13 and Table 1 show that different movement data exhibit varying spatial and temporal scales and granularities. This indicates that movement patterns can occur at different scales and granularities, which are domain-specific and task-oriented. For example, humans may exhibit flow clusters or moving clusters within small spatial extents (e.g. city blocks) and short temporal intervals (e.g. minutes to hours). Ships and animals can form similar patterns but across broader spatial regions and over long periods of time (e.g. days to weeks). Our framework can accommodate such variations across domains, because it cognizes patterns

Table 4. Summary of the three movement datasets from different domains.

Domain	Object type	Dataset description	Avg. sampling rate (s)	Spatial coverage	Duration	Number of moving objects	Number of trajectory points
Urban mobility	Humans/vehicles	GPS tracks in Beijing Sixth Ring (Geolife (Zheng <i>et al.</i> 2010))	7.01	Urban area (~2267km ²)	~5 years	182	782, 7491
Marine transportation	Ships	AIS records from the China Sea	710.89	Regional maritime zone	1 month	1838	107, 5488
Wildlife ecology	Animals (Birds)	Animal daily trajectories (Movebank (Stienen <i>et al.</i> 2023))	473.5	Cross-country regions	1 year	99	363, 7037

through a bottom-up and multi-level abstraction of movements rather than absolute spatiotemporal scales or data granularities. In contrast, baseline frameworks acknowledge such differences in scale and granularity but lack an explicit mechanism to generalize movement pattern cognition across domains with varying data characteristics.

5. Types of movement patterns that can be observed across different domains

Various movement patterns exist, and even the same types of movement patterns can emerge across different domains. Table 5 presents 32 concrete examples illustrating how typical movement patterns manifest in the urban, marine, and wildlife domains. Both our framework and the baseline frameworks can effectively characterize diverse real-world movement behaviors. However, our framework improves and extends the definitions of certain movement patterns, making them more applicable to complex and heterogeneous scenarios. For example, the convergence pattern in our framework is not limited to the strict intersection of movement paths at a fixed location with narrowly defined azimuth and temporal thresholds. Instead, it accommodates diverse converging behaviors, such as moving objects approaching a shared area from different and varying directions, especially in constrained environments, such as road networks or natural terrain. This is particularly helpful in modeling and discovering potential group events, which can significantly enhance situational awareness and support timely decision-making in applications such as emergency/disaster response and crowd management. In addition, our framework covers broader and more nuanced movement patterns than baselines. It successfully captured 100% of the 32 representative real-world domain examples in urban mobility, marine transportation, and wildlife ecology. In contrast, the coverage rates of the four baseline frameworks were 18.75% (Laube *et al.*), 31.25% (Andrienko *et al.*), 62.5% (Dodge *et al.*), and 21.88% (Li *et al.*), respectively. These results underscore the superior comprehensiveness and generalizability of our framework.

The comparison analysis of movement patterns across domains effectively validated the comprehensiveness and capability of our framework to accommodate movement data at various scales and support diverse application scenarios.

7. Discussion

7.1. Advantages of the proposed movement pattern classification framework

In this subsection, we discuss the key advantages of the proposed movement pattern classification framework while highlighting how it answers the questions raised in the introduction.

First, it offers a more comprehensive and systematic taxonomy of movement patterns. It unifies and extends existing classification schemes by integrating their core ideas and addressing their limitations. For example, we adopted the distinction between primitive/basic and compound patterns from Dodge *et al.* (2008). Guided by first principles thinking, we categorized basic patterns along three fundamental dimensions by considering the motion attribute dimension rather than only the spatial and

Table 5. Application domain examples of typical movement patterns and their coverage in our framework and baseline frameworks.

No.	Instantiations in human movement	Instantiations in ship movement	Instantiations in animal movement	Constraints	Ours	Laube's	Andrienko's	Dodge's	Li's
1	Emergency responders move at stable speeds during evacuation operations.	Naval patrol vessels cruise at a steady speed in the designated sea area.	Migratory birds fly at steady speeds over long distances.	—	Constancy	Constancy	Constancy	Constancy	✓
2	Commuter buses travel between residential zones and transport hubs.	Rescue ships ferry supplies between ports and disaster zones	Seabirds shuttle between nesting islands and feeding grounds.	Identical round-trip route	Shuttle	✓	Symmetry	Symmetry	✓
3	City workers commute to offices at fixed times every weekday.	Cargo ferries operate on fixed inter-island schedules.	Wildbeests migrate seasonally between dry and wet grazing lands.	Non-identical round-trip route	Shuttle	✓	✓	✓	✓
4	Crowds gather at the stadium for the concert at a set time.	Aid ships arrive simultaneously at the disaster relief port.	Bat colonies gather in shared caves at night for safety and warmth.	—	Spatiotemporal periodicity	✓	Periodicity	Spatiotemporal periodicity	Periodic pattern
5	Evacuees from different locations converge on emergency shelters.	Multiple rescue vessels from different ports head to a common search-and-rescue area.	Predators from various ranges converge on the local prey.	Together for all the time	Synchronized concentration	✓	✓	Concentration	✓
6				Not necessarily together all the time	Asynchronous concentration	✓	✓	✓	✓
7				From different directions without changing their original directions	Convergence	Convergence	✓	Convergence	✓
8				From different directions without retaining their original directions	Convergence	✓	✓	✓	✓
9	Individuals disperse from shelters once the evacuation phase ends	Ships exit a port and head to multiple destinations after the mission.	Flocks disperse after feeding at a shared site.	In different directions without changing the original directions	Divergence	Divergence	✓	Divergence	✓

(continued)

Table 5. Continued.

No.	Instantiations in human movement	Instantiations in ship movement	Instantiations in animal movement	Constraints	Ours	Laube's	Andrienko's	Dodge's	Li's
10				In different directions without keeping the original directions	Divergence	✓	✓	✓	✓
11	Tourists walk together toward their next destination.	Naval fleets cross the strategic strait in a coordinated formation.	Grazing animals move together in herds to increase vigilance against predators.	Fixed moving objects stay within a fixed disc for at least k consecutive timestamps	Flock	Flock	✓	Flock	Flock
12				Fixed moving objects are density-connected for at least k consecutive timestamps	Convoy	✓	✓	Fixed moving cluster	Convoy
13				Moving objects are density-connected for at least k consecutive timestamps	Moving cluster	✓	✓	Varying moving cluster	✓
14				Fixed moving objects are density-connected for at least k timestamps	Swarm	✓	✓	✓	Swarm
15	Different response teams inspect overlapping areas in a city after a disaster.	Multiple fleets dock at several identical anchorages.	Bird species stop at shared rest areas during seasonal migration.	Without temporal constraints	Co-location in space	✓	Co-location in space	Co-location in space	✓
16				With temporal information	Co-location in space and time	✓	Co-incidence in space and time	Co-incidence in space and time	✓

(continued)

Table 5. Continued.

No.	Instantiations in human movement	Instantiations in ship movement	Instantiations in animal movement	Constraints	Ours	Laube's	Andrienko's	Dodge's	Li's
17	A sequence of transit hubs frequently visited during daily commutes.	A sequence of emergency docking points visited repeatedly by coast guard vessels during maritime disaster relief.	A sequence of stopover sites inhabited by birds during their annual migrations.	—	Frequent spatial sequences	✓	Repetition	Repetition	✓
18	Courier delivers follow frequently repeated urban logistics routes.	Coast guard vessels follow patrol tracks across border zones.	Ants follow pheromone trails to and from food sites repetitively.	—	Spatial trajectory clusters	✓	Repetition	Repetition	T-pattern
19	Ride-hailing trips in the city center show overlapping trajectories during festivals.	Rescue ships exhibit aggregated movement paths toward storm-impacted coastal towns.	Bird migration paths of multiple species overlap spatially and temporally.	—	Spatiotemporal Trajectory clusters	✓	Repetition	Repetition	T-pattern
20	Urban intersections with high taxi or pedestrian density during rush hours.	Areas near disaster relief hubs where rescue ships repeatedly dock over short periods.	Feeding or watering spots where multiple animal species repeatedly converge over days or weeks.	—	Spatiotemporal hotspots	✓	✓	✓	✓
21	Ride-hailing drivers reposition themselves to high-demand zones after receiving a peak-price alert.	Relief ships or drones rerouting mid-mission to respond to emergent weather windows or updated distress signals.	Predators (e.g. lions or wolves) gather in search of a chance when they detect movement or noise.	—	Temporal opportunistic	✓	✓	✓	✓
22	Schoolchildren arrive at the school gate simultaneously.	Multiple vessels arrive at a port simultaneously for unloading.	A group of predators simultaneously arrives at a feeding site.	—	Encounter	✓	✓	Encounter	✓

(continued)

Table 5. Continued.

No.	Instantiations in human movement	Instantiations in ship movement	Instantiations in animal movement	Constraints	Ours	Laube's	Andrienko's	Dodge's	Li's
23	Crowds exit a stadium from various gates after an event simultaneously.	Fishing boats simultaneously disperse to different fishing grounds after leaving the port.	A wolf pack splits up to search separate hunting zones.	—	Breakup	✓	✓	Breakup	✓
24	Security staff arrive at a concert venue ahead of the public to conduct safety checks.	A reconnaissance ship arrives at the disaster zone to deploy communication equipment before supply ships arrive.	Scout animals (e.g. elephants or wolves) explore an area in search of food or danger before the group follows.	—	Precede in time	✓	✓	Temporal relations	✓
25	Late-arriving commuters reach a transit hub after the main rush-hour crowd has passed.	A delayed medical supply boat reaches a coastal village after primary emergency response vessels have arrived.	A predator bird arrives at the feeding ground after the rest of the flock has already started feeding.	—	Lag in time	✓	lagged synchronization	Temporal relations	
26	City bus abruptly skips multiple stops, resulting in unusually long spatial gaps between recorded positions.	A rescue boat rapidly bypasses expected waypoints on its way to the affected area.	A tagged animal makes an unexpectedly large leap between habitat zones.	—	Abnormal distance gaps	✓	✓	✓	✓
27	A subway train shows missing timestamps during operation, indicating unexpected delays or sensor failures.	A rescue ship reports irregular transmission intervals due to outages.	Wildlife tracking data shows a prolonged pause in updates, suggesting the animal was immobile or the device was malfunctioning.	—	Abnormal temporal gaps	✓	✓	✓	✓

(continued)

Table 5. Continued.

No.	Instantiations in human movement	Instantiations in ship movement	Instantiations in animal movement	Constraints	Ours	Laube's	Andrienko's	Dodge's	Li's
28	A shared car stays at a location for much longer than usual, causing unexpected delays	A supply ship is significantly delayed at sea due to unreported obstacles or rerouting	A migratory animal lingers unusually long at a stopover point	—	Abnormal delays	✓	✓	✓	✓
29	Tour guides lead tourists through urban heritage sites.	A lead vessel navigates a convoy of aid ships through a narrow strait.	A lead caribou navigates migration routes with the herd following closely behind.	—	Leadership & following	Leadership	✓	Leadership	Leadership
30	A hiker goes off established trails	A search-and-rescue boat veers from the convoy to investigate an uncharted signal.	A tagged animal unexpectedly deviates from its typical migration path.	—	Spatial path deviation	✓	✓	✓	✓
31	that most hikers take to explore an adventurous route.	Emergency boats from different launch points move toward the same coastal evacuation zone.	Birds fly south during their migrations.	—	Path deviation over time	✓	✓	✓	✓
32	Cyclists in separate bike lanes heading in the same direction.			—	Similarity	Concurrence	Similarity	Concurrence	✓

temporal dimensions. These basic patterns are combined to form compound patterns that capture the multifaceted movement behaviors of moving objects. We also drew on the classification perspectives of similarity, difference, and arrangement proposed by Andrienko and Andrienko (2007) to categorize the basic patterns within each dimension. Therefore, our framework can cover a broader range of movement patterns, including not only REMO patterns (considering only motion attributes) but also movement patterns with spatial constraints.

Second, our framework distinguishes movement patterns across different levels of movement abstraction based on the proposed multi-level movement cognition model. It includes the micro atomic level for fine-grained movements; the meso episodic level emphasizing key points, segments, or regions of interest; and the macro flow level focusing on global movement dynamics. This enables the framework to adapt to different scales and granularities of movement data across domains.

Third, our classification framework is both generalizable and adaptable to various domains. On the one hand, it was developed based on fundamental dimensions intrinsic to the movements themselves, independent of domain-specific constraints (e.g. object types and semantic labels). On the other hand, the framework enables the refinement and customization of movement patterns through a set of well-defined constraints and the 5W1H1R ontology model, including both basic spatiotemporal properties and high-level semantic descriptions. This greatly enhances the ability of our framework to derive domain-specific or compound movement patterns that are commonly encountered in practice but rarely covered systematically in earlier classification frameworks.

Our movement pattern classification framework is flexible and extensible. By enumerating all possible combinations of basic patterns across different dimensions, it is theoretically possible to derive a complete set of compound movement patterns. In addition, the modular design of the movement pattern classification allows for easy expansion. New dimensions can be introduced and integrated with existing ones without disrupting the overall taxonomy of movement patterns, which is crucial for addressing emerging requirements in future research and applications.

7.2. Multi-level cognition and the mineability of movement patterns

As discussed in Section 7.1, the proposed classification framework can be applied to diverse moving objects and application domains. Movement patterns are cognized from three bottom-up levels of movement abstraction rather than fixed or absolute spatiotemporal scales or granularities. Movement patterns can always be classified and analyzed across these three levels for moving objects with different motion spaces and time lengths.

One notable aspect is the relationship between movement patterns and the scale, granularity, and resolution of movement data. Even the same type of movement pattern can be observed at different spatial and temporal scales (small to large and local to global). For example, a moving cluster can be a small group of people walking together in a park (small scale) or a fleet of ships sailing in formation on the ocean (large scale). It is crucial to recognize that spatial and temporal scales do not always

vary proportionally. A long-term pattern does not necessarily imply a large spatial scale or high flow level. For example, three ships moving in coordination for two months involves a long temporal span and wide spatial coverage, but can still reflect atomic-level movement behavior. Conversely, nationwide flows of individuals by air are considered short-term patterns with a large spatial scale and extent at the flow level. There is no universal or absolute size for spatial or temporal scales in our classification framework. The relationship between space and time is domain-specific and must be flexibly modeled according to the application context.

Since movement data resolution and granularity affect the movement patterns that can be detected, we propose the concept of the mineability of movement patterns. In this study, it is defined as the feasibility and accuracy with which movement patterns can be extracted from movement data, given their spatiotemporal scale and data resolution. Analogous insights can be found in studies on human movement predictability. Researchers have shown that predictability varies with the spatial and temporal resolution of the data (Song *et al.* 2010, Smolak *et al.* 2022). Similarly, in movement pattern mining, the spatiotemporal resolution and granularity of movement data influence the mineability of movement patterns. In general, lower-resolution data reduces the mineability of certain movement patterns, because the spatial or temporal details necessary for identifying patterns, especially those at the atomic level, may be lost. For example, moving clusters can be detected more clearly with sufficient temporal and spatial granularity; however, they may become blurred or entirely lost when data resolution is reduced. However, this influence is not strictly linear or uniform across pattern types and levels.

At the micro atomic level, movement patterns generally rely heavily on high-resolution data that capture fine-grained movement details. However, such details may also amplify noise and outliers, thereby interfering with the detection of movement patterns. At the meso episodic level, which focuses on key points or regions, a moderate resolution is typically sufficient, although higher granularity can improve the mineability of movement patterns. In contrast, movement patterns at the macro flow level are less sensitive to changes in resolution. Lower-resolution data may still be sufficient to capture overall trends.

Finally, we should note that data resolution is a relative concept. High or low resolution depends on the spatial and temporal characteristics of the moving objects being studied. For example, a 5-minute sampling interval may be a high resolution for tracking maritime ships but too coarse for studying the movement of urban vehicles. Therefore, the choice of data resolution should consider the movement characteristics of the object types and the level of the movement patterns being analyzed.

7.3. Application and implementation of the movement pattern classification framework

In this subsection, we further discuss how our framework can be applied in practice. This study contributes to several aspects. First, the 5W1H1R model introduces multiple cognitive dimensions to support a comprehensive understanding of movement patterns. This enables researchers to interpret movement patterns not only in terms of spatiotemporal properties or statistical descriptors but also in terms of intentions, processes, and results, thus providing a richer analytical lens. Second, the multi-level

model enables a bottom-up understanding of movement patterns across the atomic, episodic, and flow levels. This supports not only the identification of point-level movement behaviors (e.g. moving clusters and abnormal turning points), but also key point/region-based and flow-based movement behaviors (e.g. hotspot stops and aggregated flows). Third, the classification framework built on these two cognition models provides a systematic taxonomy and description of movement patterns. Researchers can use this framework to guide the discovery of new movement patterns and to interpret and compare movement patterns across domains.

Although our classification framework does not directly implement movement pattern mining algorithms, it offers a conceptual foundation for guiding the mining and analysis of movement patterns in different domains. On the one hand, it can guide the design of specific movement pattern mining algorithms for specific domains and assist in developing general algorithms for specific movement patterns across domains. On the other hand, there is a growing interest in the development of unified algorithms for various movement patterns. Recent studies have attempted to integrate multiple existing detection modules or have already applied deep learning to classify movement patterns. With the development of deep learning and large language models (LLMs), researchers in various fields have begun to explore multimodal models for various tasks, such as remote sensing multimodal models. This provides unprecedented opportunities, inspiring the development of unified movement pattern mining models by combining deep learning and LLM techniques. To achieve this goal, a comprehensive cognition and taxonomy of movement patterns is essential. Our framework can serve as such a cognitive and ontological foundation for the development of multimodal or universal movement pattern detection methods.

8. Conclusions and future work

This study developed a unified framework for the systematic characterization of movement patterns. In our framework, the 5W1H1R principle is proposed to recognize movement patterns from a horizontal perspective across multiple dimensions. A multi-level movement cognition model was developed from a vertical perspective to recognize movement patterns at three distinct levels of abstraction. Subsequently, a novel and extended taxonomy of movement patterns, guided by first principles thinking and combinatorics, was established and validated in three application domains. We summarized five categories of movement constraints (spatial, temporal, motion attributes, moving objects, and geographical environment) to further subdivide movement patterns into more granular forms and apply them to the three common movement patterns. Compared to existing movement pattern classification frameworks, our framework covers a broader range of movement patterns across different levels of movement abstraction and can adapt to varying scales and granularities. It demonstrates greater flexibility for domain-specific adaptations and future expansions.

Although movement pattern mining has emerged as a promising research direction in the field of spatiotemporal data mining, it remains relatively underdeveloped compared to other well-established spatiotemporal data mining tasks. This study adopts ideas from existing research, thereby laying a conceptual and theoretical foundation

for future advancements in this field. Our framework can guide the creation of benchmark movement datasets, the design of visual analytics tools, the exploration of movement pattern mineability, the development of multimodal movement pattern mining models, and the development of highly efficient pattern mining algorithms for streaming movement data, which will be the focus of our future work.

Acknowledgments

The authors gratefully acknowledge the comments from the editor and the reviewers.

Disclosure statement

The authors reported no potential conflict of interest.

Funding

This study was supported by funds of the National Key Research and Development Program of China (Grant No. 2022YFB3904203), National Natural Science Foundation of China (42430110, 42271462, 42471487), the Science and Technology Innovation Program of Hunan Province (2024AQ2026), Hunan Provincial Natural Science Foundation of China (2025JJ20038, 2024JJ1009), the Scientific Research Fund of Hunan Provincial Education Department (24A0023), Key Laboratory of Target Cognition and Application Technology(2023-CXPT-LC-005) and the Funds of Open Projects of Hunan Engineering Research Center of Geographic information security and application (No. HNGISA2024001).

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Data and codes availability statement

This research is theoretical in nature and does not involve any algorithm code that needs to be shared. The three datasets used for empirical validation of the proposed classification framework and the corresponding codes are publicly available on 'figshare.com', with the identifier at the public link: <https://doi.org/10.6084/m9.figshare.29820188>.

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